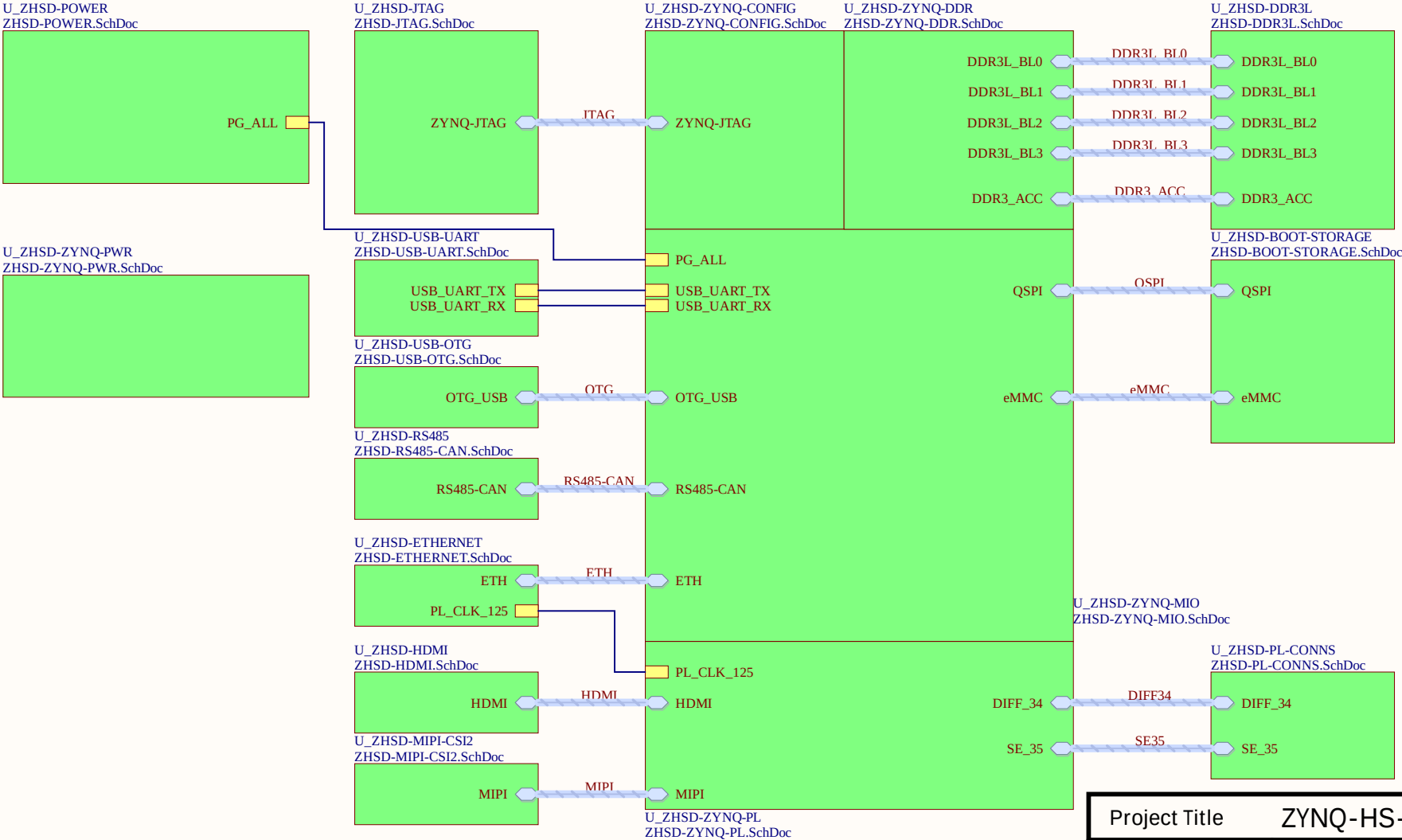


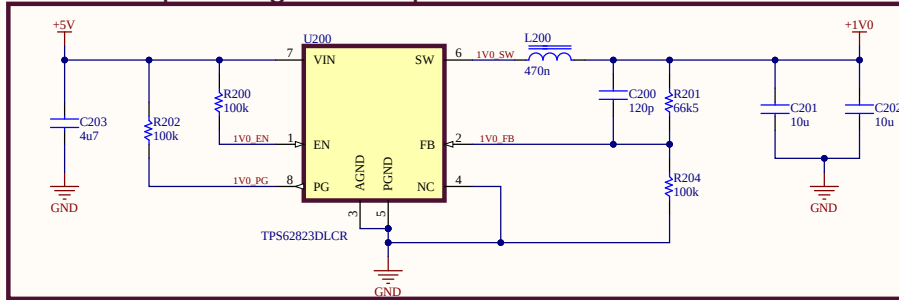
ZYNQ-7000 HIGH SPEED DESIGN



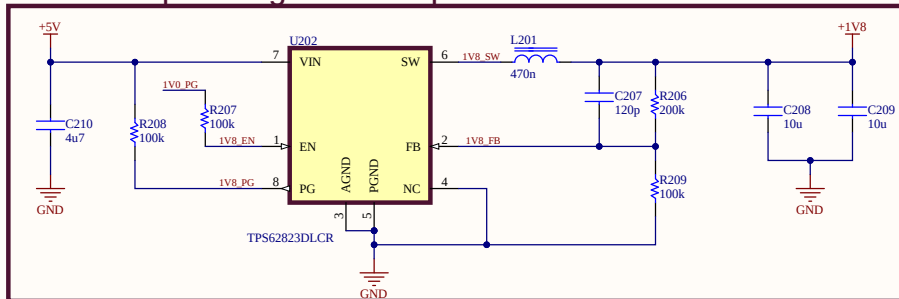
Project Title		ZYNQ-HS-DEV.PrjPcb	
Drawn By		Sheet Name	
Uğur KELEŞ		ZHSD-Block.SchDoc	
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ZHSD POWER SYSTEM

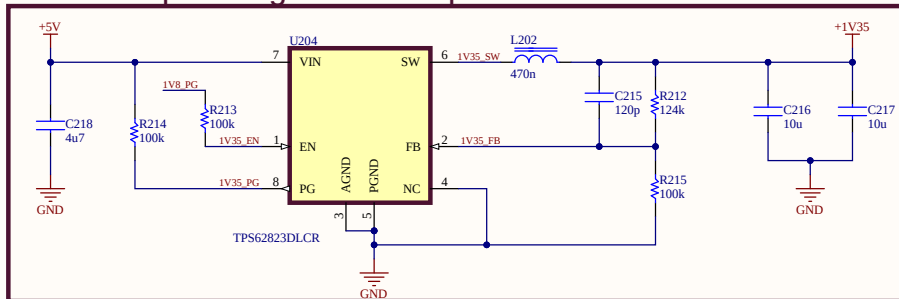
Power Sequencing - 1V Output



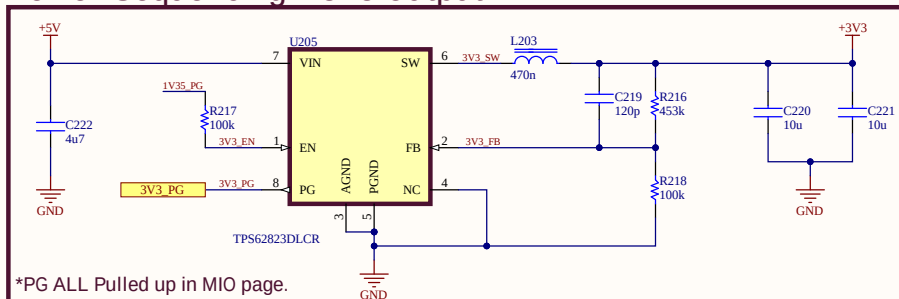
Power Sequencing - 1V8 Output



Power Sequencing - 1V35 Output

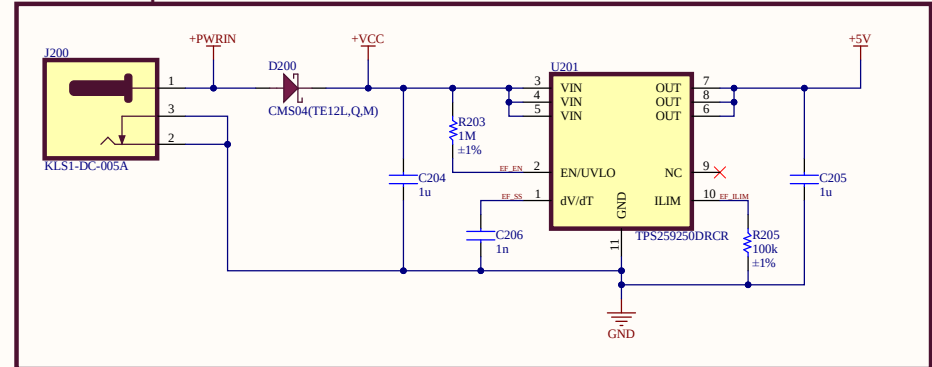


Power Sequencing - 3V3 Output

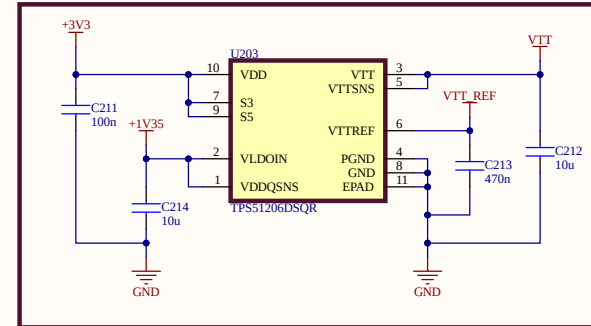


*PG ALL Pulled up in MIO page.

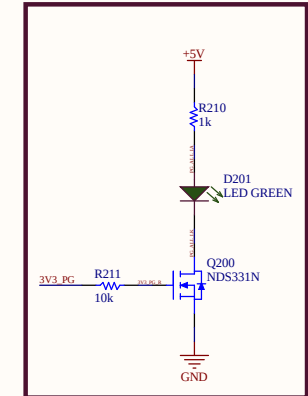
Power Input & EFuse



VTT



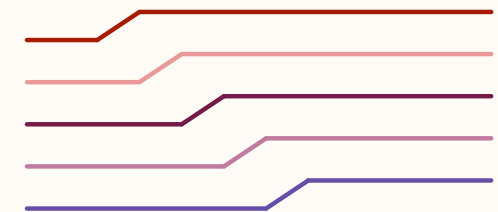
PG Led



Power Sequence

*Zynq-7000 SoC DC and AC Switching Characteristics DS187

- +1V0 VCCPINT, VCCINT, VCCBRAM
- +1V8 VCCPAUX, VCCPLL, VCCO_MIO1, VCCAUX
- +1V35 VCCO_DDR
- +3V3 VCCO_MIO0, VCCO(PL Banks)
- +VTT DDR3L Reference Voltage

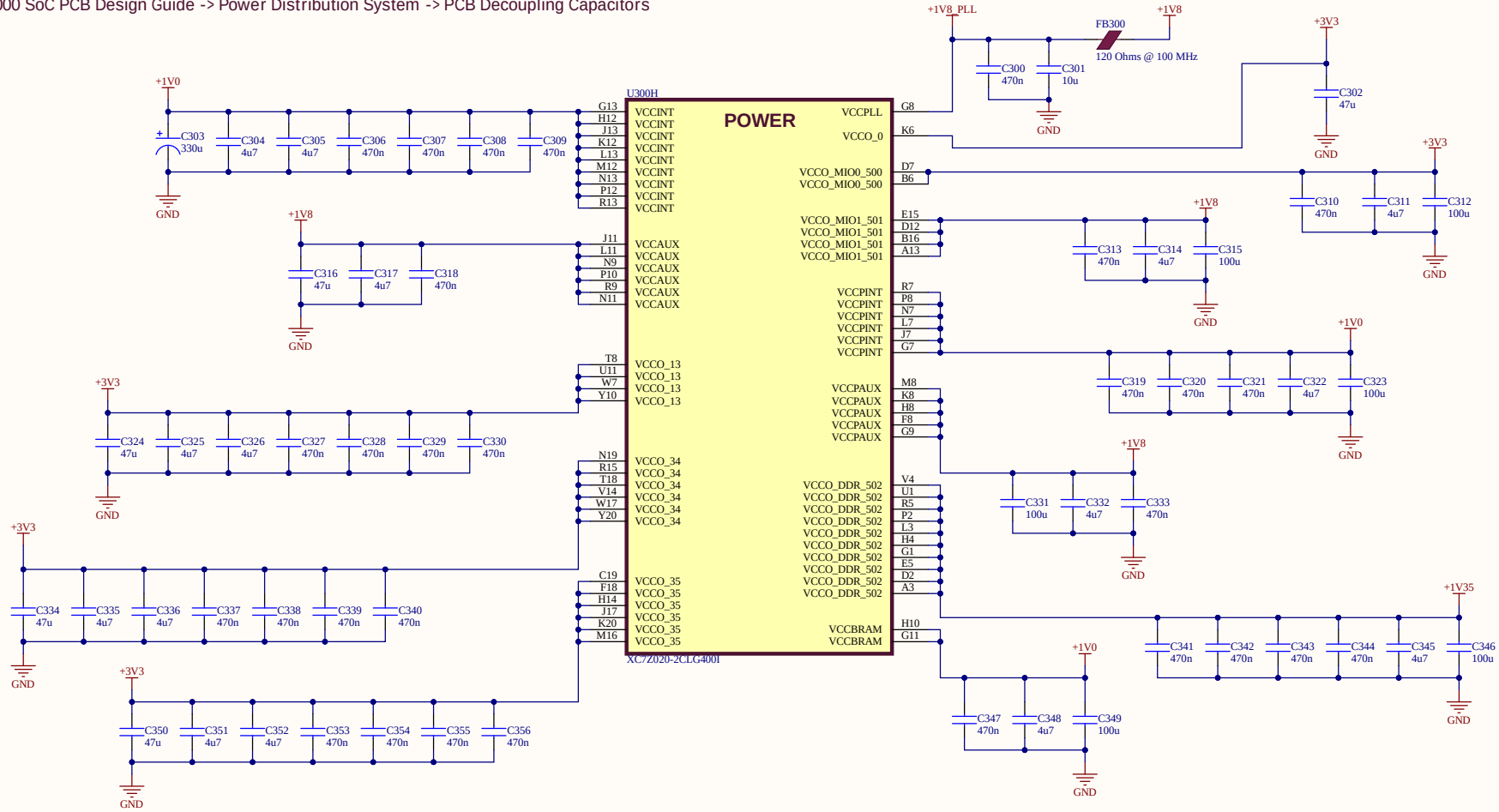


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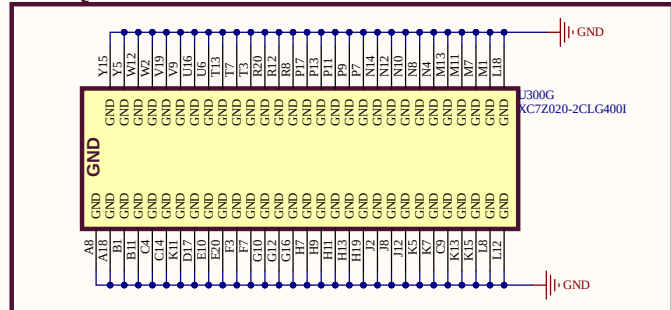
ZHSD ZYNQ POWER DECOUPLING

ZYNQ 7000 Power Decoupling

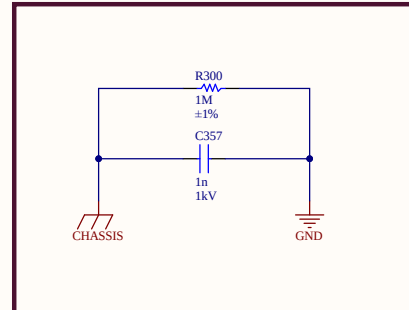
*Zynq-7000 SoC PCB Design Guide -> Power Distribution System -> PCB Decoupling Capacitors



ZYNQ 7000 - GND



ZHSD CHASSIS-GND

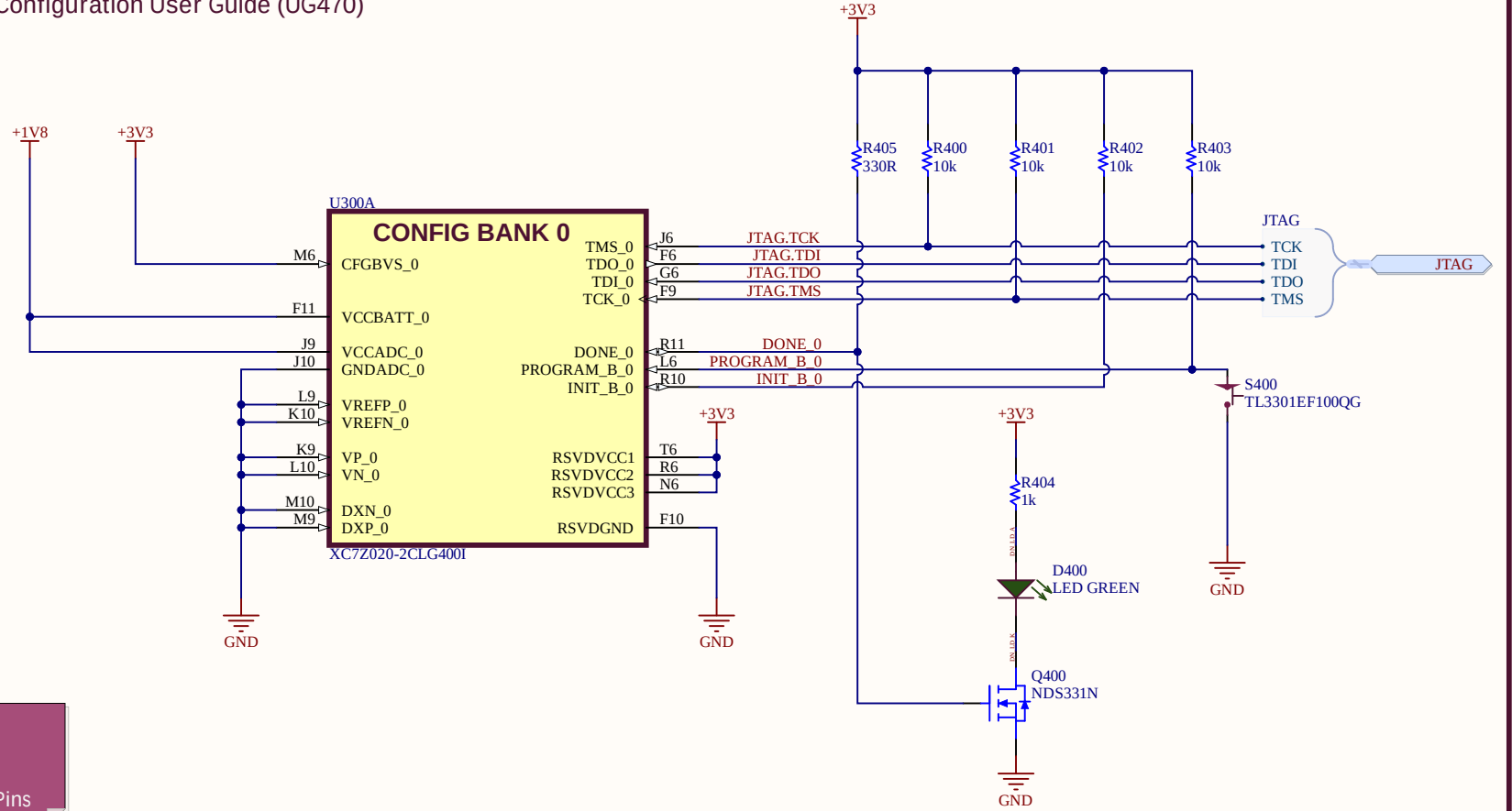


Project Title			ZYNQ-HS-DEV.PrjPcb		
Drawn By Uğur KELEŞ		Sheet Name ZHSD-ZYNQ-PWR.SchDoc			
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ZHSD ZYNQ CONFIG

ZYNQ-7000 Config Bank 0

*7 Series FPGAs Configuration User Guide (UG470)

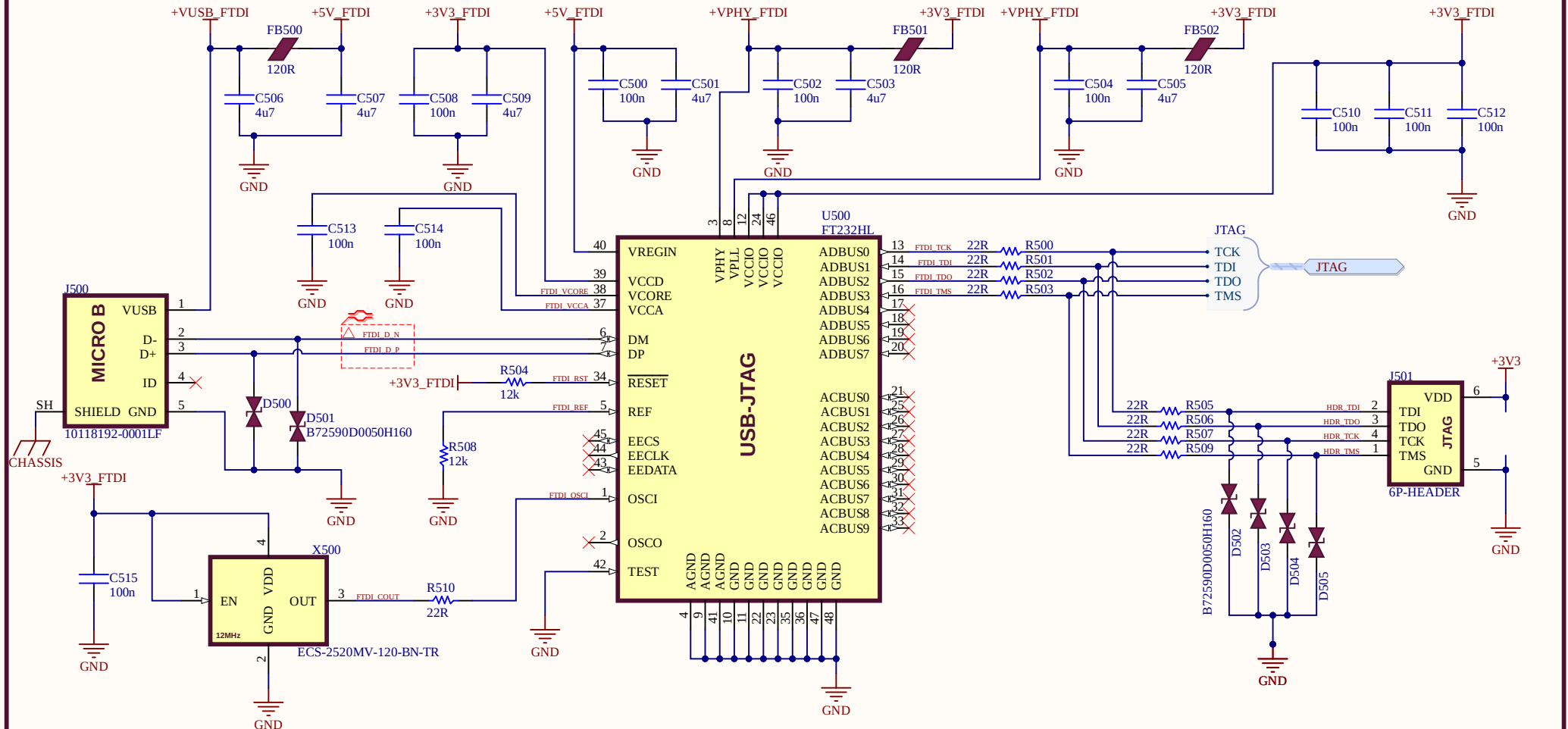


UG480 p9 xADC
UG933 p68 DXP/N
UG865 p13 RSVD
UG470 p20 Other Pins

Project Title		ZYNQ-HS-DEV.PrjPcb	
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Uğur KELEŞ		ZHSD-ZYNQ-CONFIG.SchDoc	
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USB-JTAG

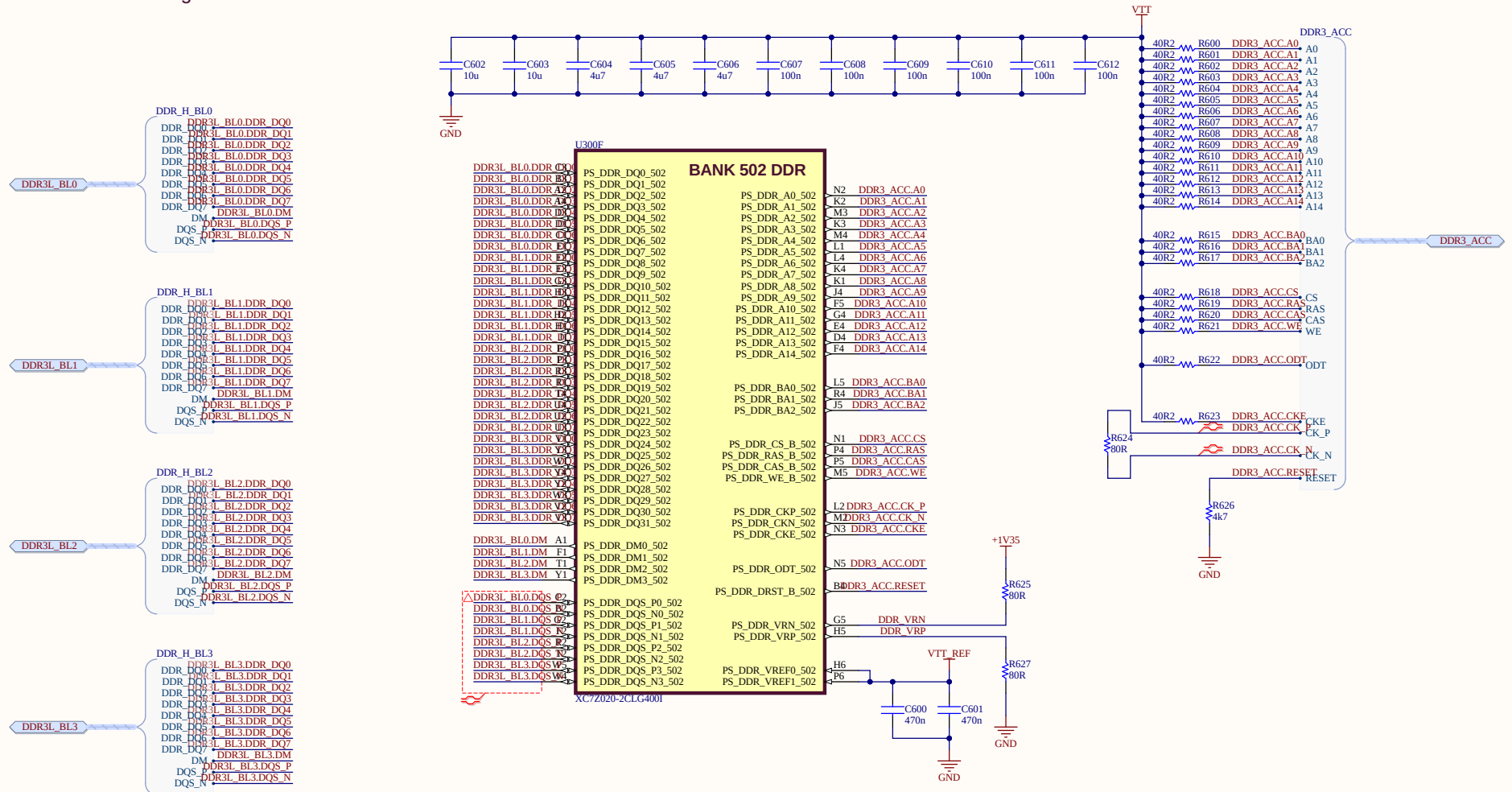
Project Title		ZYNQ-HS-DEV.PrjPcb	
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ZHSD ZYNQ DDR3L

ZYNQ-7000 DDR BANK 502

*Zynq-7000 SoC PCB Design Guide UG933



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DDR3L IC



Sheet Name
ZHSD-DDR3L.SchDoc

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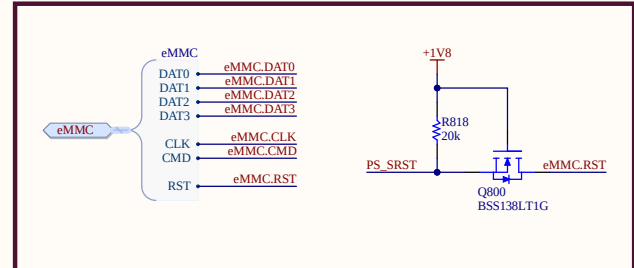
A

B

C



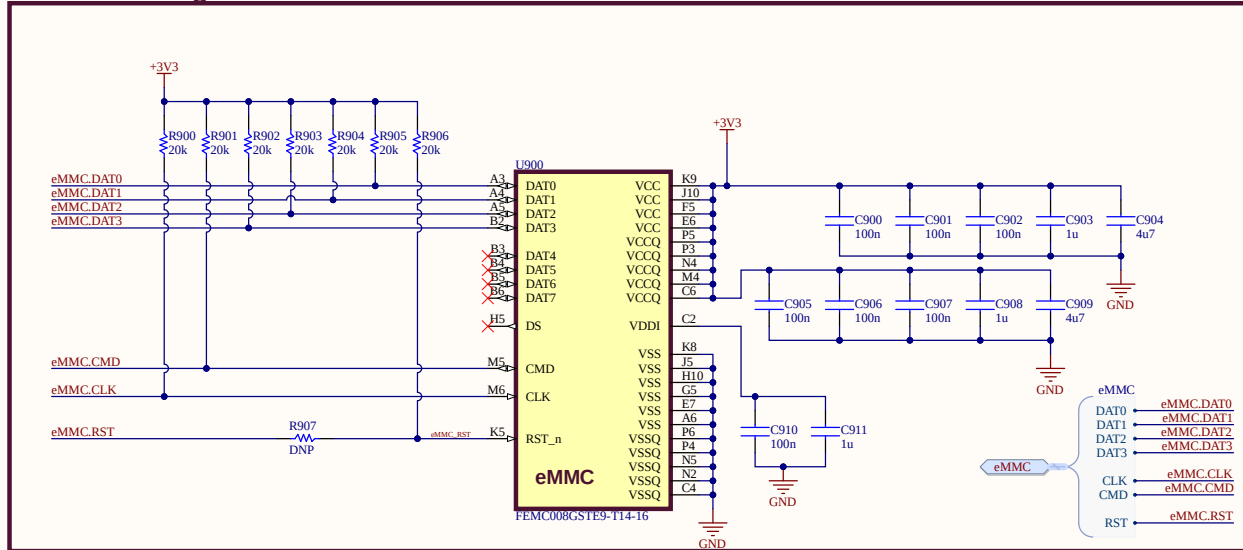
MC Harness & Level Translation



1	2	3	4	5	6	7	8
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ZHSD BOOTING & STORAGE

eMMC Storage



ZYNQ Boot Modes Reference

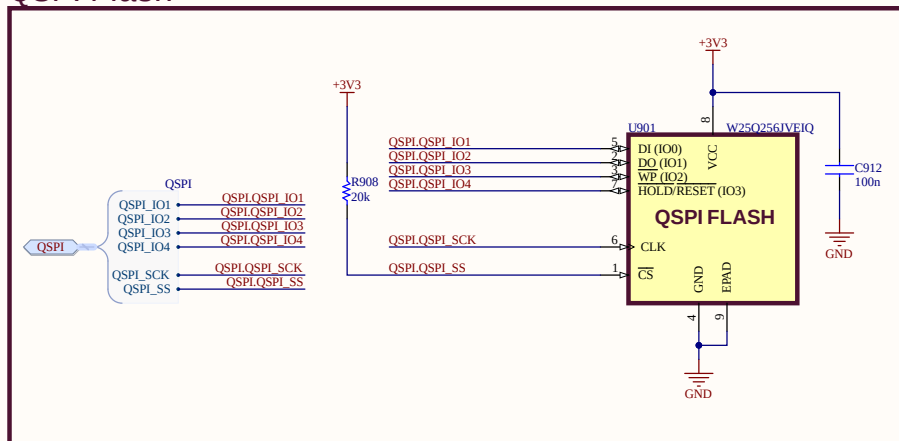
Table 6-4: Boot Mode MIO Strapping Pins

Pin-signal / Mode	MIO[8]	MIO[7]	MIO[6]	MIO[5]	MIO[4]	MIO[3]	MIO[2]
	VMODE[1]	VMODE[0]	BOOT_MODE[4]	BOOT_MODE[0]	BOOT_MODE[2]	BOOT_MODE[1]	BOOT_MODE[3]
Boot Devices							
JTAG Boot Mode; cascaded is most common ⁽¹⁾			0	0	0		JTAG Chain Routing ⁽²⁾ 0: Cascade mode 1: Independent mode
NOR Boot ⁽³⁾			0	0	1		
NAND			0	1	0		
Quad-SPI ⁽³⁾			1	0	0		
SD Card			1	1	0		
Mode for all 3 PLLs							
PLL Enabled		0	Hardware waits for PLL to lock, then executes BootROM.				
PLL Bypassed		1	Allows for a wide PS_CLK frequency range.				
MIO Bank Voltage ⁽⁴⁾							
	Bank 1	Bank 0	Voltage Bank 0 includes MIO pins 0 thru 15.				
2.5 V, 3.3 V	0	0	Voltage Bank 1 includes MIO pins 16 thru 53.				
1.8 V	1	1					

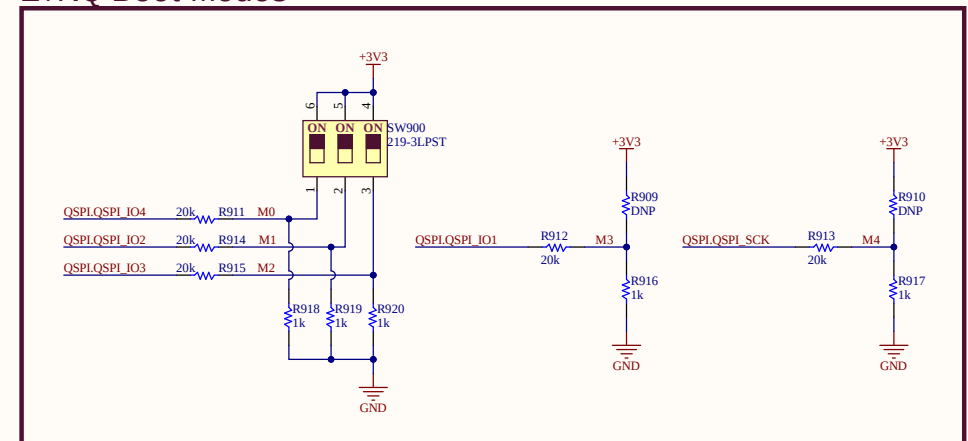
Notes:

1. JTAG cascaded mode is most common and is the assumed mode in all the references to JTAG mode except where noted.
2. For secure mode, JTAG is not enabled and MIO[2] is ignored.
3. The Quad-SPI and NOR boot modes support execute-in-place (this support is always non-secure)
4. Voltage Banks 0 and 1 must be set to the same value when an interface spans across these voltage banks. Examples include NOR, 16-bit NAND, and a wide TPIU test port. Other interface configuration may also span the two banks.

QSPI Flash



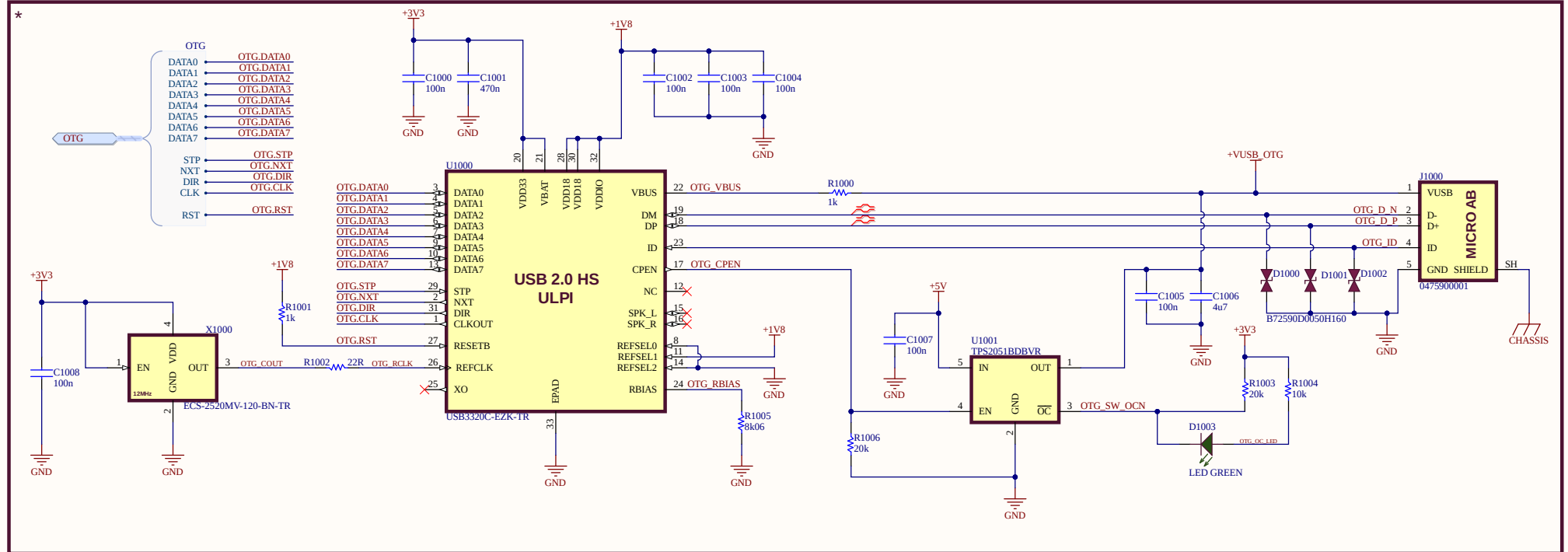
ZYNQ Boot Modes



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ZHSD USB OTG 2.0 ULPI

USB OTG PHY & Micro AB Connector



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Drawn By Uğur KELEŞ	Sheet Name ZHSD-USB-OTG.SchDoc		
Date 1.02.2026	Sheet 10 / 17	Sheet Size A3	

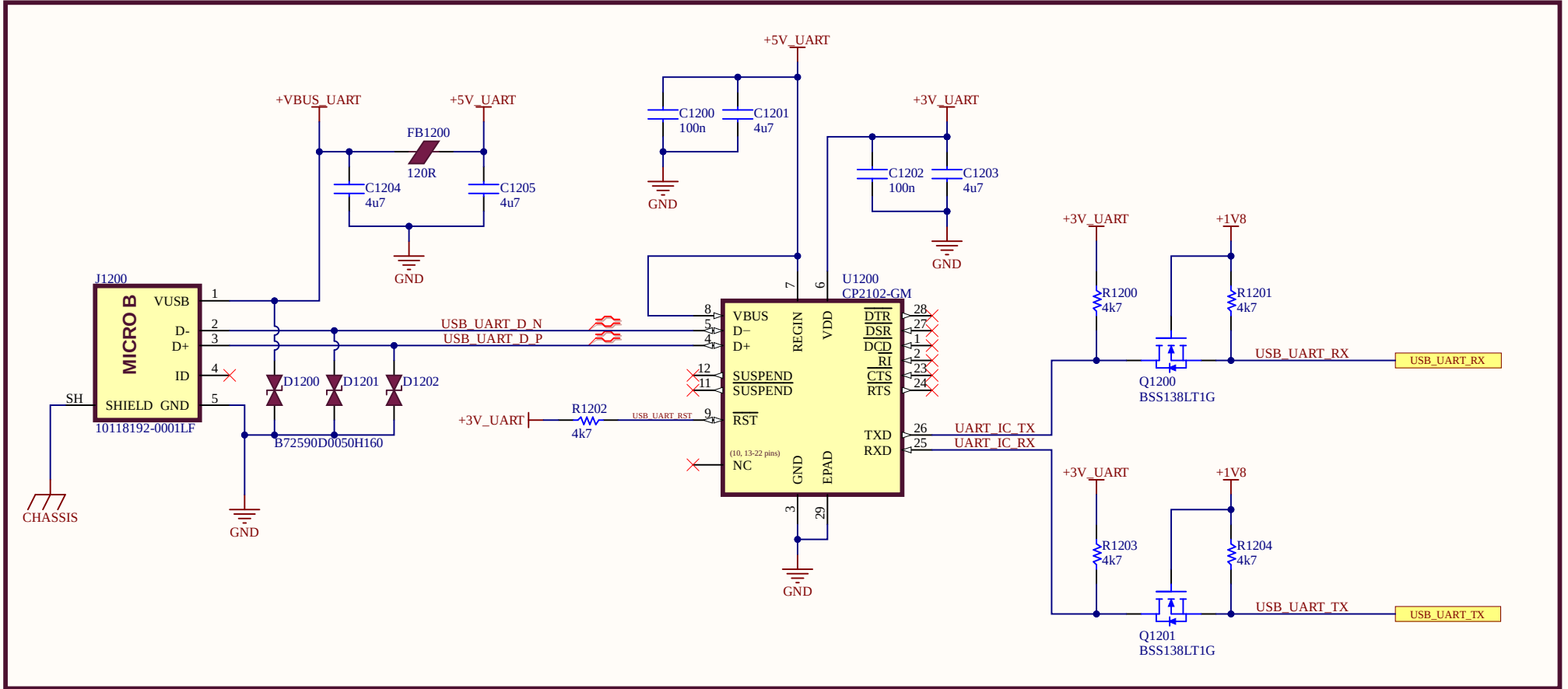
A

B

D

USB-UART INTERFACE

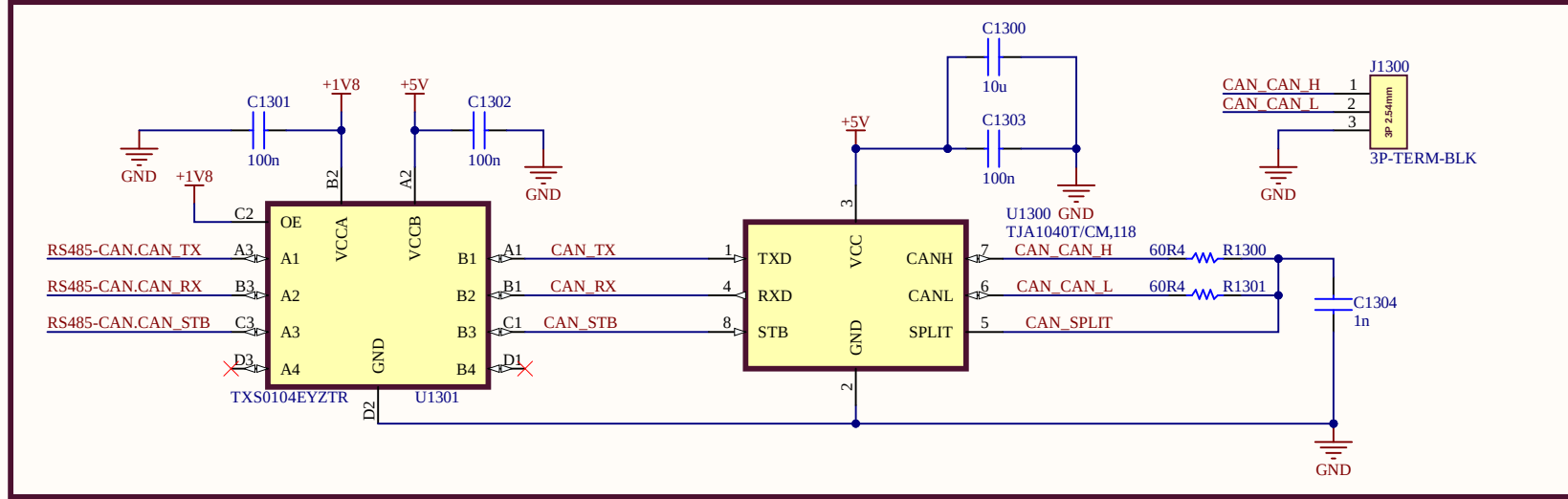
USB-UART



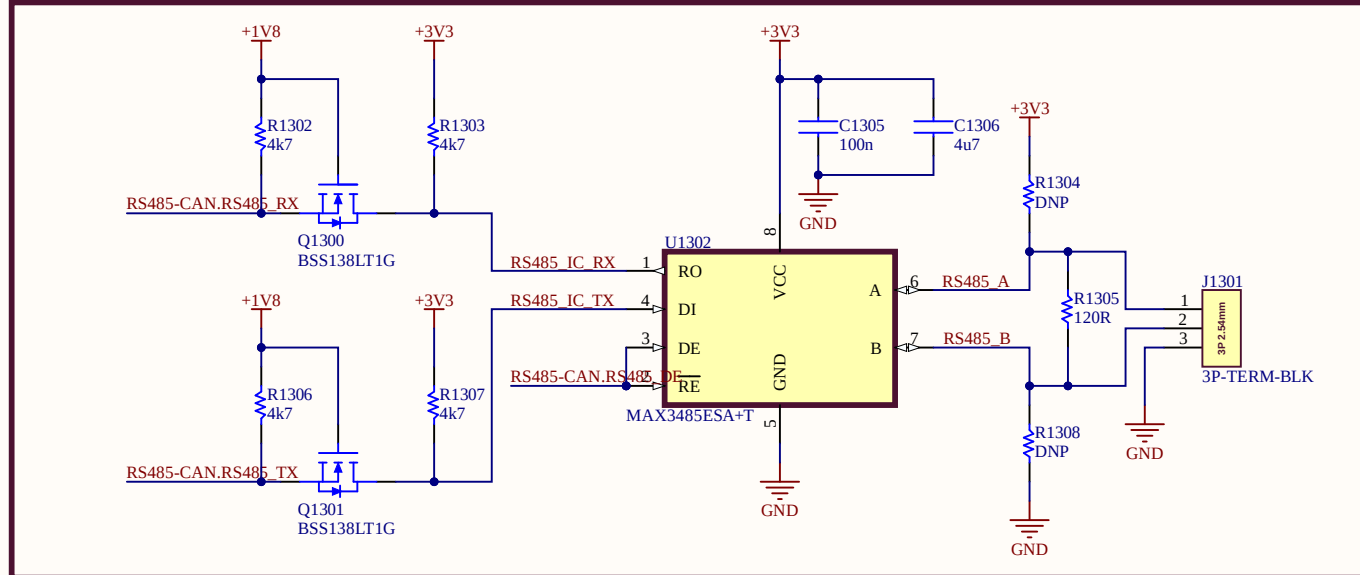
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Drawn By Uğur KELEŞ		Sheet Name ZHSD-USB-UART.SchDoc	
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RS485 & CAN INTERFACES

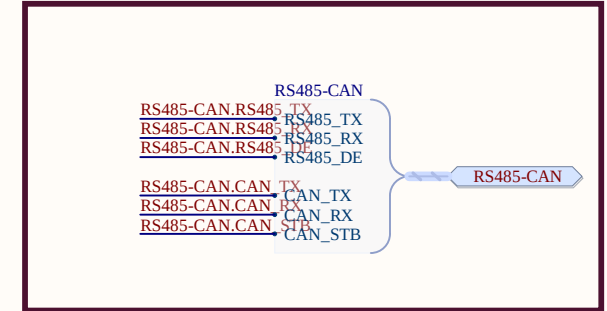
CAN Trceiver & Level Translation



RS485 Transceiver

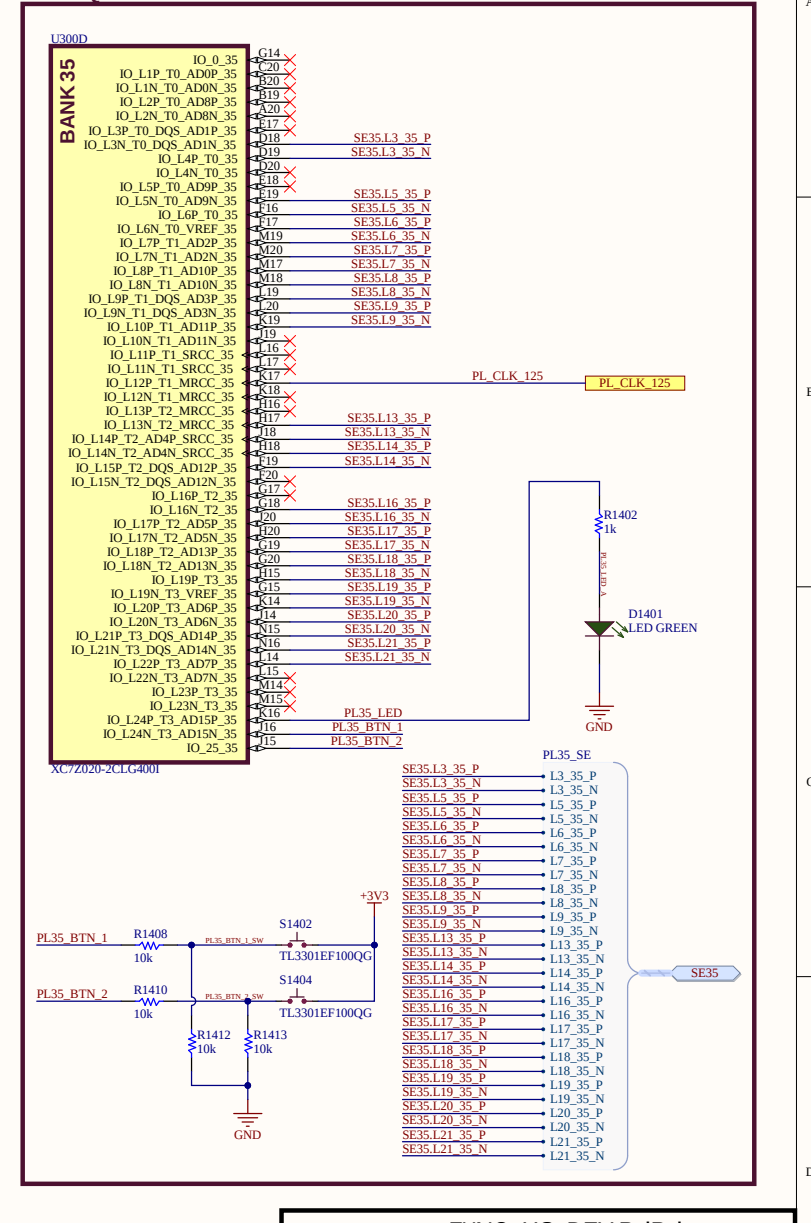


RS485 & CAN Harness



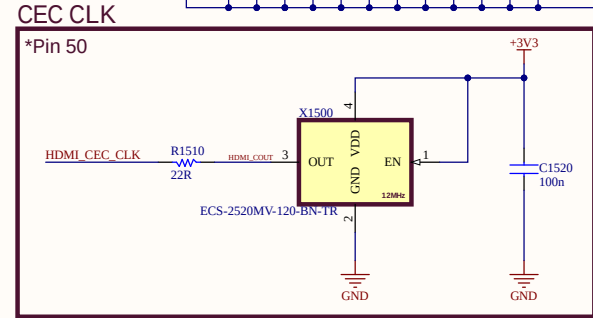
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Uğur KELEŞ		ZHSD-RS485-CAN.SchDoc	
Date	1.02.2026	Sheet	13 / 17
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ZYNQ-7000 PL Bank 35



Project Title		ZYNQ-HS-DEV.PrjPcb	
Drawn By Uğur KELEŞ		Sheet Name ZHSD-ZYNQ-PL.SchDoc	
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HDMI Transmitter and Connector

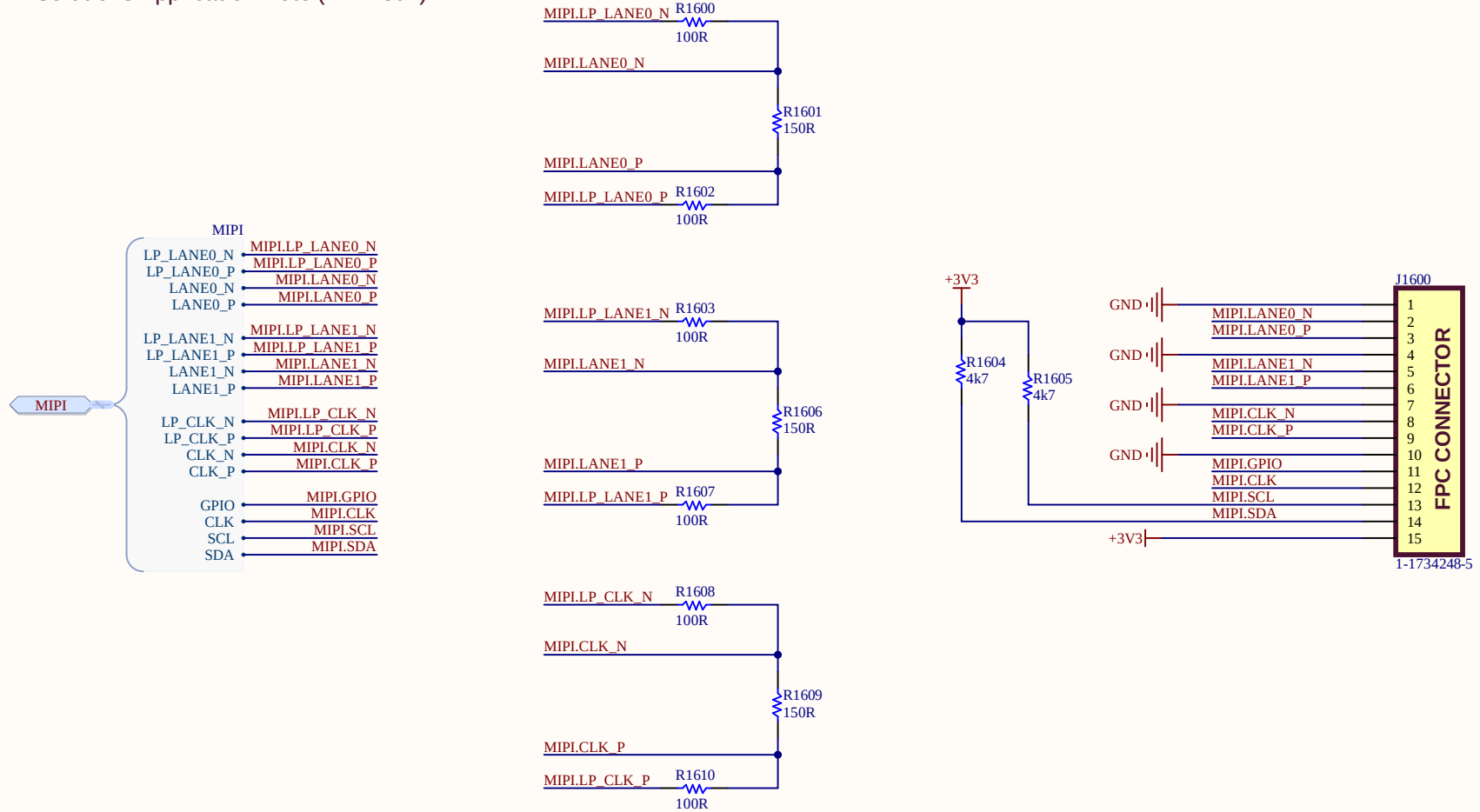


Project Title		ZYNQ-HS-DEV.PrjPcb	
Drawn By Uğur KELEŞ		Sheet Name ZHSD-HDMI.SchDoc	
Date	1.02.2026	Sheet	15 / 17
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ZHSD MIPI INTERFACE

MIPI CSI-2

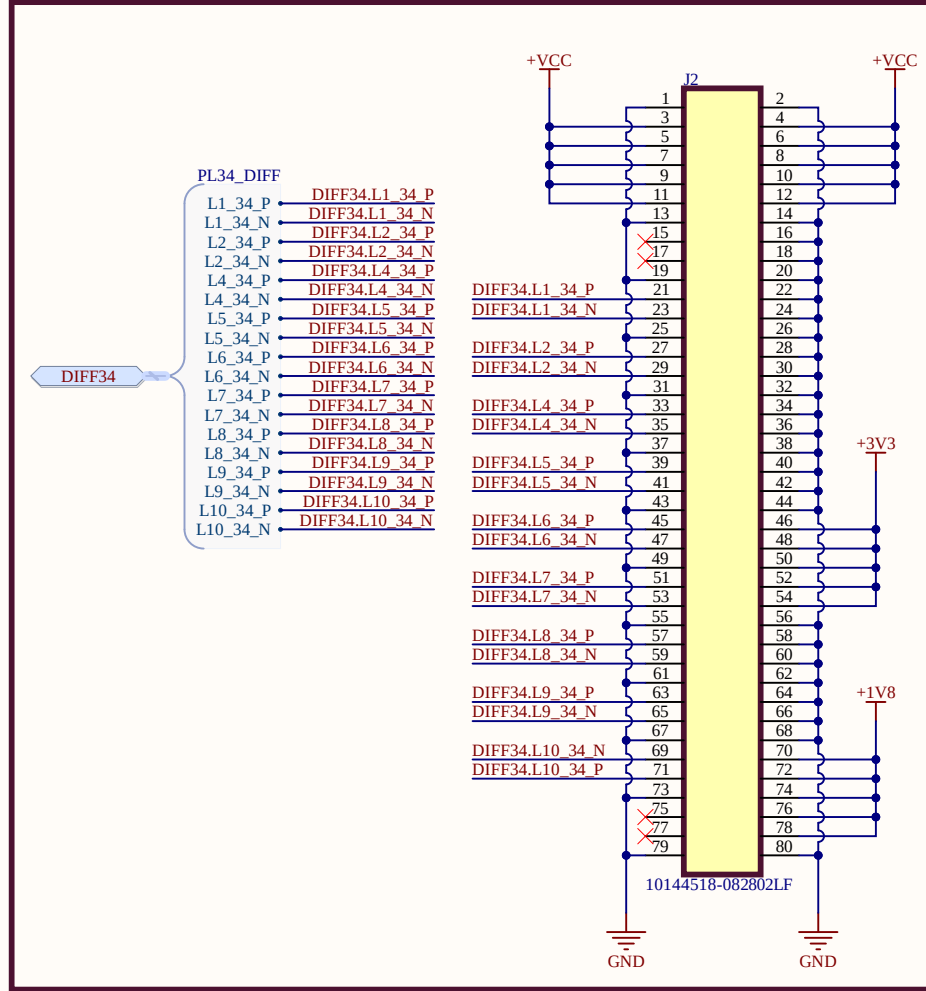
*D-PHY Solutions Application Note (XAPP894)



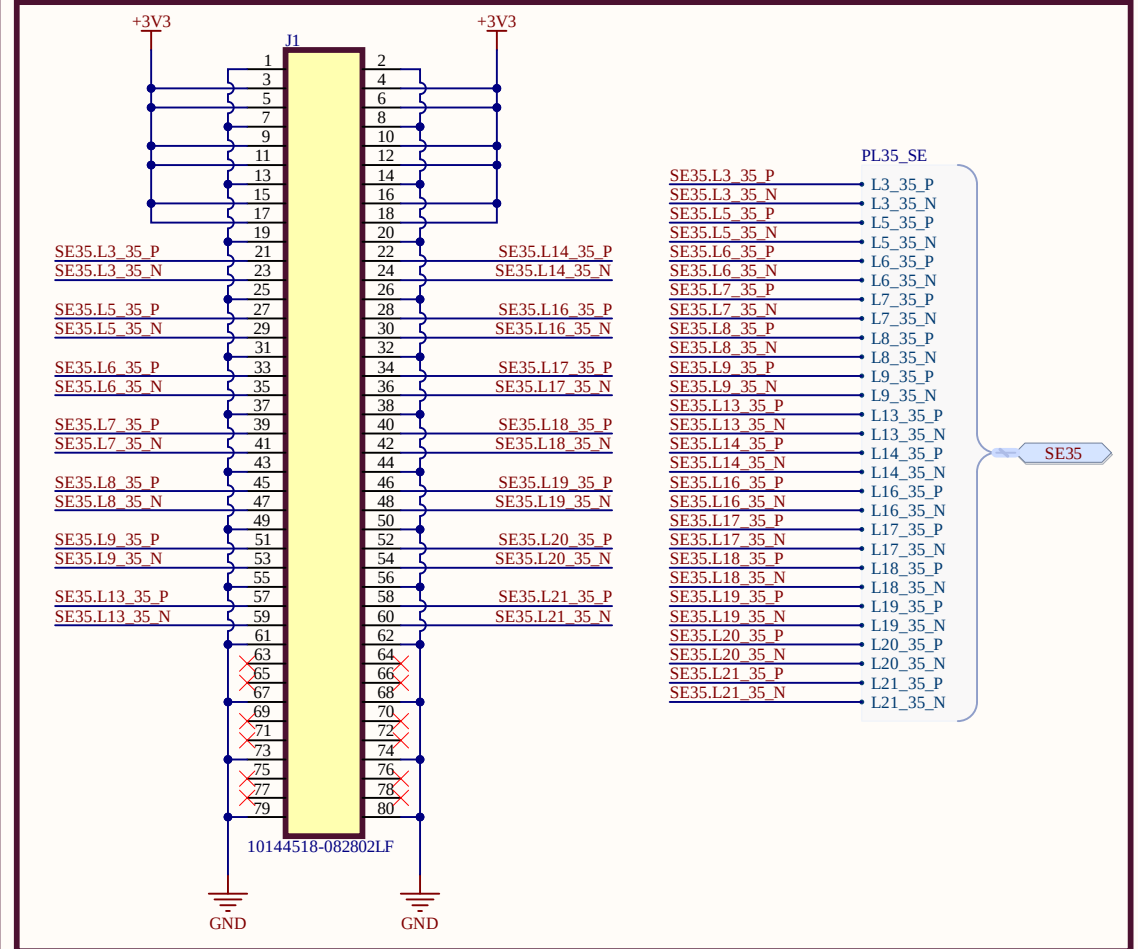
Project Title		ZYNQ-HS-DEV.PrjPcb	
Drawn By		Sheet Name	
Uğur KELEŞ		ZHSD-MIPI-CSI2.SchDoc	
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ZHSD PL CONNECTIONS

PL34 Differential Signals



PL35 Single Ended Signals



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